

Supplementary material

Text S1

The model

Ecosim is a tool for dynamic simulations based on the Ecopath model, an instantaneous image of the ecosystem at a given time [1-3]. Ecopath models account for the biomass of each functional group of species, their diet composition, production per unit of biomass (P/B), consumption per unit of biomass (Q/B), mortality from predators and fishing, accumulation of biomass and net migration. The principle behind this ecosystem modelling approach is that, on a yearly basis, biomass and energy are conserved. The proportion of the mortality of each group that is accounted by the model (fishing, predation, biomass accumulation, migration) is called ecotrophic efficiency (EE). The P/B is considered equal to total mortality under equilibrium condition [4] since production and losses would be equal. The total mortality (Z) is thus computed as the sum of fishing mortality (F) and natural mortality (M) which includes mortality by predation.

Ecosim uses a system of differential equations to describe changes in biomass and flows with the system by accounting for change in predation and fishing

$$\frac{dB_i}{dt} = g_i \sum_j Q_{ji} - \sum_j Q_{ij} + I_i - (M_{0,i} + F_i + e_i) B_i$$

eq. 1

where g_i is the net growth efficiency; Q_{ji} and Q_{ij} are the consumption rate of group j by group i and the consumption of group i by group j respectively; I_i the immigration in t/km^2 ; $M_{0,i}$ the instantaneous non-predatory natural mortality; F_i the fishing mortality; and e_i the emigration rate. Ecosim can incorporate the multiple life history stages (stanzas) for species having ontogenic changes in trophic and habitat preferences. The stanzas are linked and their respective production and consumption per unit of biomass (P/B_i and Q/B_i respectively, per year), and growth are calculated from a baseline estimate for the leading group (the adults in our case). It is assumed that growth for each stanza follows a von Bertalanffy growth curve, and that the survivorship is stable through ages.

The estimation of consumption of prey i by predator j (Q_{ij}) at each time step is based on the foraging arena theory [2,5] and calculated as:

$$Q_{ij} = (a_{ij} v_{ij} B_i B_j) / (2v_{ij} + a_{ij} B_j)$$

eq. 2

where a_{ij} is the rate of effective search for prey i , v (vulnerability) is the rate of exchange between the vulnerable and invulnerable prey biomass pools. The estimate of a_{ij} is obtained by solving equation 2 using parameters from the Ecopath model and conditional on the value of v (default value=2). Low vulnerability ($1 > v_{ij} < 1.5$) implies a donor-control or type II functional response, while a large value implies that a change in biomass of the predator will cause a corresponding change in the mortality rate of its prey. Model fitting is achieved by estimating vulnerability values that minimizes the sum of squares of differences between model predictions and times series of biomass and catch.

Data

Note: Except for the groups designating fish of a particular family or one species, fish groups names are typically composed of first, their habitat preference (coastal, pelagic, shelf), and second, their size (small (S), medium (M) or large (L)) (Table S1).

Fish

Biomasses of demersal fish, octopus and cephalopods were derived from the annual trawl survey and estimated using a delta general linear model (delta-GLM) accounting for strata, year, and catchability coefficients for groups of taxa, following the method described in [6,7]. Pelagic fish biomass were obtained from yearly acoustic surveys estimates carried out by the Norwegian vessel N/O Dr. Fridtjof Nansen, covering the north-western coast of Africa (region) starting before 1995, and the vessel El Awam, covering the Mauritanian coast starting in 2003. The biomass of sardine and mackerel, available only at the scale of the region, were downscaled assuming that the biomass present in Mauritania was proportional to the ratio of Mauritanian catch to that of the whole region:

$B_{\text{maur}} = B_{\text{region}} * C_{\text{maur}} / C_{\text{region}}$. In addition, the biomass of horse mackerel estimated from the Norwegian vessel for Mauritania was downscaled using the average biomass estimate from the Mauritanian survey. Such estimates are potentially unreliable given the variable migrations linked to variations in upwelling strength. Consequently, the time series of biomass of mackerel, sardine, and anchovy were considered doubtful and non-representative, and were not used to fitting the model. The (negative) biomass accumulation rates (/year) were calculated from the biomass trends observed for the various exploited groups before 1991 (see [6]) for fish and cephalopods (Table 1)

The residence time of migratory species in the study area was derived from knowledge of their migrations and timing of their fishery. Small pelagics were generally present 8 months of the year [8], meagre 6 months [9] and mullets 5 months [10]. Residence times were used to scale down the biomass present in the study area. Biomasses of fish present in the Banc d'Arguin were estimated from the trawl surveys Amrigne in 2000-2007. These estimates, although incomplete and probably biased because of the gear limitation in such a coastal environment, were added to the shelf biomass estimate but they are often insignificant compared to estimates for the rest of the shelf.

Fish Q/B values were preferably estimated with the empirical equation from [11]:

$$\log_{10}(Q/B) = 7.964 - 0.204\log_{10}(W_{\infty}) - 1.965T' + 0.083A + 0.532h + 0.398d \quad \text{eq. 3}$$

where Q/B is the yearly food consumption of a fish population as a percentage of its biomass, W_{∞} , the mean asymptotic (or maximum) weight (g) of the fish in the population in question, $T' = 1000/\text{Kelvin}$ (Kelvin = °C + 273.15) is the mean habitat temperature, and d (=1 for detritivores), carnivores being identified when both h and d equal 0, A=aspect ratio of the tail. An alternate estimate was obtained with:

$$Q/B = 106.37 * 0.0313(1000/273.15 + T) * W_{\infty}^{-0.168} * 1.38P_f * 1.89H_d \quad \text{eq. 4}$$

where the food type ($P_f = 1$ for carnivores and $H_d = 1$ for detritivores and herbivores). We preferred the results of equation 3 as equation 4 generally produces higher (too high) Q/B estimates. For selacians and fish for which we did not have sufficient information, the ratio P/Q was fixed at 0.15 and 0.2 respectively.

Natural mortality (M) of fish was derived from the empirical model of Pauly [13]:

$$\log_{10}(M) = 0.333 - 0.246 * \log_{10}(L_{\infty}) + 0.744 * \log_{10}(K) + 0.01 * T \quad \text{eq. 5}$$

where K and L_{∞} (cm) refer to the curvature and asymptotic (total) length parameters of the von Bertalanffy growth function, and T is the mean annual water temperature in Celsius (19.5 °C). Alternative estimates were taken from Hoenig's empirical equation [14] for species for which the longevity is available:

$$\ln(M) = 1.44 - 0.982 * \ln(\text{max age}) \quad \text{eq. 6}$$

using the maximal observed age found in Fishbase. The lower of the two estimates was kept.

Fish diets were preferably derived from studies from the region or elsewhere in similar habitats, as found in the literature and in Fishbase, and local qualitative information from local fisheries biologists (Table S2). The composition of each fish groups is listed in Table S1.

Fishing data

Catches for the artisanal and the industrial (demersal and pelagic) fleets were obtained from the Mauritanian database. There is no information on discards for any of the fleets. Catches classified as 'other fish' were allocated to the groups Coastal M and Shelf M (25% each) and the remaining 50% to the other demersal fish in proportion to the amount already reported. Catches for small pelagics, octopus (*Octopus vulgaris*) and ethmalosa (*Ethmalosa fimbriata*, group coastal M) were obtained from the estimates of the Fishery Committee for the Eastern Central Atlantic Working Group [15]. Effort trends consisted in the number of vessels for artisanal and industrial demersal fleets [6] while the average gross registered tonnage (GRT) was used for the pelagic fleet [16]. During the study period, effort increased 5 times for the artisanal fleet and 2 times for industrial pelagic and demersal fleets (Table S3).

Mammals and birds

The marine mammal group includes only species mostly feeding in the study area, ignoring oceanic species that feed marginally in the area [17] (Table S4). Monk seals biomass (*Monachus monachus*) was taken from a previous model of the Mauritanian coast [18]. Biomasses for other marine mammals were obtained from the density estimates derived from global databases and habitat characteristics for the north-western African coast [17], assuming similar densities in Mauritania as in the region, and Q/B ratios were calculated using an empirical equation based on body weight [19] (Table S4). P/B ratios were set at 0.02 /year for killer whales, 0.036 for beaked whales [19] and 0.05 for all other species collectively [17] (Table S4). Diet compositions were obtained from [20] and completed with qualitative information found in other publications [21,22]. There are no time series for birds and mammals.

Coastal birds are either breeders or migratory species spending part of the year in Mauritania, feeding in coastal waters or on muddy flats in the Banc d'Arguin (Table S5). The abundance of breeding birds was calculated from counts of breeding pairs assuming that the total population amounted to 3 times the breeding pairs [23]. Waders were counted in the winter in the Baie d'Arguin, an area carrying about the same density of waders as the entire Banc but 18 times smaller [24]. The authors estimated the possible numbers of birds in the Baie at 1.4 to 2 fold the actual counts. Hence, the number of waders counted in the Baie was multiplied by 25.5 to obtain the abundance for the Banc d'Arguin (Table S5). The ration in kg/day was estimated from an empirical equation [25]. Diet compositions were derived from both qualitative and quantitative studies found in the literature. Natural mortality was obtained from compilations found in the literature for 7 species [26-28]. The resulting natural mortality (M, 0.28/year) is consistent with a weighted average maximal age of 17 years for the 20 species for which the information was available.

Cephalopods

Cephalopods were divided in two groups, octopus and other cephalopods. Octopus mortality was estimated at 0.1/month for a life span of about 1.5 year and adults dying massively after spawning [29]. Thus, M was estimated at 1.2 /year and F at 0.64 based on the ratio C/B. Q/B was estimated by the model using a P/Q value of 0.3. Other cephalopods were assumed to have similar natural mortality and P/Q ratio as octopus, while fishing mortality (L/B) amounted to 0.43. Diet compositions were derived from qualitative and quantitative studies for *Octopus vulgaris* [30], *Loligo forbesi* and *L. vulgaris* [31-33] and *Illex illecebrosus* [34].

Plankton

Zooplankton on the shelf was obtained from transect sampling outside the Banc d'Arguin, which means covering the northern part of the ZEE [35]. The mesozooplankton comprises copepods, cladocerans, gastropod and lamellibranch larvae, dinoflagellates, etc. Their biomass is estimated at 68 t/km² using an average depth of 66 m, conversion ratio AFDW:DW of 0.904:1 and DW:WW 1:0.186 from [36], and assuming that densities are similar in the southern part of the ZEE. The macrozooplankton includes euphausiids, mysids, chaetognaths, salps, hydrozoa, fish larvae, etc. Their biomass was estimated at 4.21 t/km² using taxa specific conversion ratios from AFDW to WW [36]. There was no information for zooplankton in the Banc, so these biomasses were left for the model to estimate using an EE of 0.8. P/B and Q/B values for mesozooplankton were set at 24 /year and 112

/year based on [37]. This value of P/B is lower than estimates compiled for *Pseudocalanus* in various north hemisphere seas (52 (32-73) per year in [38]). Herbivorous macrozooplankton were estimated to ingest a maximum of about a third of their body weight per day [35] or 110 /year. P/B and Q/B values for macrozooplankton were set at 4.3 and 17 based on a compilation of estimates for northern seas [39]. Mesozooplankton were assumed to feed mainly on phytoplankton while macrozooplankton's diet was based on qualitative information found in the literature (see [39]).

Benthic fauna

The biomasses of shelf macrobenthos were obtained from two transects (north and south) estimates crossing the shelf from the Banc to the limit of the shelf [40], using taxa specific conversion ratios from AFDW to WW [36]. Polychaetes were the main group found throughout the plateau and no gradient in biomass was found with depth. The authors attributed the higher biomass resulting from their study compared with that of the southern shelf found in the literature to the effect of the high productivity around the upwelling in the north (Cape Blanc).

In the Banc d'Arguin, the biomass of benthic invertebrates was estimated only for tidal flats that spans about 500 km² [41]. Shrimps, sampled with push nets in both the intertidal and subtidal areas of the Banc, may be underestimated as the area sampled is small and small shrimps were not included in the analysis [42]. Densities of shrimps, molluscs and worms were estimated by assuming that the biomass outside the subtidal habitats was equal to that of the intertidal (Table S6). The violonist crab (*Uca tangeri*) and the gastropod *Senilia senilis* were both assumed to be restricted to tidal flats. Small crustaceans (amphipods, isopods, etc) were thought to be underestimated given the sampling gear and thus their density was assumed to be a minimal estimate for the entire area of the Banc. As a first approximation, meiobenthos biomass was assumed to amount to about 6% of the macrobenthos biomass, the observed proportion for the shelf, but their biomass was increased to balance the model. P/B values were obtained from the literature [43-45] and P/Q values (=gross efficiency) from [43].

Primary production

Phytoplankton species composition in the Banc d'Arguin is different from that of the upwelling area where diatoms are abundant [46]. Crude estimates of phytoplankton biomass for the Banc d'Arguin were obtained from 2 studies, one covering the Baie du Lévrier in 1973 and the other the inshore waters [46]. The productivity in mgC/m²/d were converted in wet weight using a C:WW ratio of 1:9 [47] and assuming a P/B of 100. The average of these two estimates amounted to 31 t/km² in the Banc, ignoring the very low biomass on tidal flats [24]. The primary production for the shelf, 2,549 mgC/m²/day, was derived from SeaWifs and obtained from the Sea Around Us project web site (<http://www.seaaroundus.org/eez/478.aspx>). Using the same C:WW conversion ratio, the biomass amounted to 83.7 t/km².

The biomass of seagrass beds of tidal and subtidal habitats was derived from Vermaat et al. [48] using DW:WW ratio of 0.199:1 [36] for a total of 1707 g/m² in the Banc. The Banc covers 43% of the littoral area between 5 and 20 m deep and seagrass beds outside the Banc were assumed to be half as dense as in the Banc. Thus, the Banc was assumed to contain 72% of the seagrass biomass. The average above ground production was estimated at 2 AFDW g/m²/d, or 6921 gWW/yr using the same ratios as above, resulting in a P/B of 4.1/year.

Balancing and fitting the model

Using the input values, Ecopath solves simultaneous linear equations and estimates the missing parameters, often the Ecotrophic Efficiency (EE) value. The balancing process is done manually by checking inconsistencies in data, adjusting biomasses, P/B ratio, and diet composition, starting with parameters that were deemed less reliable. As such, diet compositions are often modified on the account of seasonal and individual variation and sampling error. Overestimates of the proportion of rare prey in the diet of an abundant predator is a common source of excessive mortality. P/B values are often overestimated and the P/Q ratio is sometimes too low because of Q/B overestimates (e.g. [39]).

Time series of biomass, catch and fishing effort by fleet were used to fit Ecosim temporal simulations for the period 1991-2006. The model was driven by fishing effort (Table S3) and fit to the observed biomass and catch (Listed in Table S7). The fitting occurred in two recursive steps aiming at decreasing the sum of squares between the predicted and observed biomass and catch. The model was first explored to identify the most sensitive predator-prey relationships and second, the vulnerability values were estimated using the formal non-linear search procedure. We used the knowledge gained in the manual fitting to start the search procedure with different initial parameters to avoid being caught in local optimal solutions.

In addition, the biomass time series of mackerel, catfish and sardine were not considered (weight=0 on a scale of 0 to 1) to fit the model because their time series were not deemed representative of the population trends. The biomass time series of coastal S and shelf L crustaceans were given a weight of 0.5 because of their large variability and the little number of species the time series covered compared with the number of species included in these groups. The search for vulnerability values was carried out only for functional groups with time series, excluding sardines.

The upwelling indices for the study period was obtained from IMROP (Institut Mauritanien de Recherches Océanographiques et des Pêches) and used to modify the production of the shelf phytoplankton group.

Balancing and fitting results

Balancing this model required to change the diet of several groups. Predation had to be decreased on most juveniles groups, Coastal M, cephalopods, shelf soles, and the 2 groups of macrozooplankton. The predation was transferred in large part to adult groups, sardinelles, sardine, shelf S, coastal S, and the 2 groups of worms and mesozooplankton. The values of P/Q was low for several pelagic species and thus the P/Q ratio was fixed at 0.2, leaving Ecopath to estimate a more appropriate value of Q/B for sardine, horse mackerel and mackerel. The biomass of shelf soles, octopus, BA meiobenthos and BA crustaceans were increased. The biomass of sardinelles was obviously too large and the EE too small so the biomass was decreased from 24 to 18 t/km².

Several explorative trials at fitting the model to time series running from 1991 to 2006 led to the conclusion that the index of upwelling did not improve the fit to demersal or pelagic species. Changes in primary production induced by the index of upwelling introduced more noise in biomass trends for demersals (except for octopus), increasing the sums of squares for these groups. The observed biomass trends of sardine, sardinelles, mackerel, and horse mackerel exhibit different trajectories during the study period as they are function of several factors such as the strength of the upwelling, physical conditions and their impact on plankton dynamics, and fish habitat preferences [8]. These signals can also be confounded by changes in the position of their geographical centre of gravity. For instance, sardines increased off the Western Sahara (north of Mauritania) as a result of a southward displacement of their centre of distribution in the 1970s [8]. Finally, the upwelling index for the study period is rather flat, except for increases in 1991-1994 and 1997-1998. Hence, it is not surprising that a model with no detail on phytoplankton and mesozooplankton cannot resolve the dynamics of each small pelagic species. Our objectives being to assess the role of the Banc d'Arguin, we did not attempt to fit the small pelagic component trends any further, expecting the model to explain mostly the dynamics of the demersal component. Fitting was improved by taking into account the negative biomass accumulation (except for shelf soles). P/B ratios had to be decreased for hake, octopus, cephalopods. The sum of squares of the fitted model is 280 compared to the initial 423 when all vulnerability values were set at 2.

Mixed trophic impact (MTI)

Direct and indirect interactions within the ecosystem were analyzed using the Mixed Trophic Impact (MTI) routine of Ecopath, which assesses the relative impact of a slight increase in abundance of any group on the biomass of other groups in the food web [2]. The MTI index, scaled from -1 to 1, was calculated for every group of the model. The net cumulative impact Q_{ij} of a predator j on a prey i is calculated as:

$$q_{ji} = DC_{ji} - Q_{ji}/Q_j$$

where DC_{ji} is the proportion of the prey i in the predator's diet (the positive effect of the prey), while Q_{ji}/Q_j is the negative effect, Q_{ji} is the quantity eaten by predator j of prey i and Q_j is the total consumption by predator j . To this direct impact are added all indirect impacts of the predator on the prey calculated as the sum of all indirect paths calculated as the product of the Q_{ij} values in each paths [1,49].

Table S1. Composition of fish functional groups.

N	Group	Species
3	Meagre	<i>Argyrosomus regius</i>
5	Mullets	<i>Mugil cephalus</i> ; <i>Mugil capurrii</i> ; <i>Liza aurata</i> ; <i>Liza falcipinnis</i> ; <i>Liza ramada</i> ; <i>Mugil cephalus</i> ; <i>Liza dumerili</i>
6	Pelagic L	<i>Orcynopsis unicolor</i> ; <i>Euthynnus alletteratus</i> ; <i>Sarda sarda</i> ; <i>Auxis thazard thazard</i> ; <i>Auxis rochei rochei</i> ; <i>Pomatomus saltatrix</i> ; <i>Trichiurus lepturus</i> ; <i>Lepidopus caudatus</i> ; <i>Scomberomorus tritor</i> ; <i>Mola mola</i> ; <i>Sphyrna guachancho</i> ; <i>Sphyrna sphyraena</i> ; <i>Lichia amia</i> ; <i>Coryphaena hippurus</i> ; <i>Elops lacerta</i>
7	Mackerel	<i>Scomber japonicus</i>
8	Sardine	<i>Sardina pilchardus</i>
9	Sardinelles	<i>Sardinella maderensis</i> ; <i>Sardinella aurita</i> ; <i>Engraulis encrasicolus</i>
10	Horse mackerels	<i>Trachurus trachurus</i> ; <i>Trachurus trecae</i> ; <i>Caranx rhonchus</i>
11	Coastal selacians	<i>Dasyatis centroura</i> ; <i>Dasyatis pastinaca</i> ; <i>Gymnura altavela</i> ; <i>Leptocharias smithii</i> ; <i>Mustelus mustelus</i> ; <i>Myliobatis aquila</i> ; <i>Pteromyiaeus bovinus</i> ; <i>Rhinobatos cemiculus</i> ; <i>Rhizoprionodon acutus</i> ; <i>Torpedo marmorata</i> ; <i>Zanobatus schoenleinii</i> ; <i>Rhinobatos rhinobatos</i> ; <i>Dasyatis margarita</i> ; <i>Dasyatis margarita</i> ; <i>Raja miraletus</i> ; <i>Rhinoptera bonasus</i> ; <i>Rhinoptera marginata</i> ; <i>Dasyatis marmorata</i>
12	Coastal M	<i>Solea solea</i> ; <i>Cynoglossus senegalensis</i> ; <i>Solea senegalensis</i> ; <i>Cynoglossus canariensis</i> ; <i>Pegusa lascaris</i> ; <i>Cynoglossus monodi</i> ; <i>Synaptura lusitanica</i> ; <i>Synaptura lusitanica nigromaculata</i> ; <i>Synaptura cadenatii</i> ; <i>Pegusa triophthalma</i> ; <i>Psettodes belcheri</i> ; <i>Psettodes bennettii</i> ; <i>Dicentrarchus punctatus</i> ; <i>Aluterus schoepfii</i> ; <i>Balistes punctatus</i> ; <i>Balistes carolinensis</i> ; <i>Parapristipoma octolineatum</i> ; <i>Galeoides decadactylus</i> ; <i>Pentanemus quinquarius</i> ; <i>Mullus barbatus</i> ; <i>Eucinostomus melanopterus</i> ; <i>Drepane africana</i> ; <i>Scorpaena elongata</i> ; <i>Ethmalosa fimbriata</i> ; <i>Sarotherodon melanotheron melanotheron</i> ; <i>Brachydeuterus auritus</i> ; <i>Stephanolepis hispidus</i> ; <i>Lethrinus atlanticus</i> ; <i>Chaetodipterus lippei</i> ; <i>Syngnatus typhle</i> ; <i>Spaeroides marmoratus</i> ; <i>Sphaeroides spengleri</i> ; <i>Gobius niger</i> ; <i>Pomatoschistus marmoratus</i> ; <i>Pomatoschistus microps</i> ; <i>Solitas gruvelli</i> ; <i>Atherina lopeziana</i>
13	Coastal S	<i>Pseudolithus senegalensis</i> ; <i>Pseudolithus typus</i> ; <i>Pseudolithus senegalensis</i>
14-15	Croakers	<i>Plectorhinchus mediterraneus</i> ; <i>Sparus aurata</i> ; <i>Diplodus puntazzo</i> ; <i>Diplodus sargus sargus</i> ; <i>Diplodus sargus cadenatii</i> ; <i>Diplodus vulgaris</i> ; <i>Diplodus bellottii</i> ; <i>Dentex canariensis</i> ; <i>Dentex gibbosus</i> ; <i>Pagrus pagrus</i> ; <i>Pagrus caeruleostictus</i> ; <i>Pagrus auriga</i> ; <i>Pagellus bogaraveo</i> ; <i>Pagrus pagrus</i> ; <i>Lithognathus mormyrus</i>
16-17	Seabreams	<i>Arius heudelotii</i> ; <i>Arius latiscutatus</i> ; <i>Arius parkii</i>
18-19	Catfish	
20	Shelf selacians	<i>Galeorhinus galeus</i> ; <i>Paragaleus pectoralis</i> ; <i>Raja undulata</i> ; <i>Oxynotus centrina</i> ; <i>Raja straeleni</i> ; <i>Squalus blainville</i> ; <i>Symphodus bailloni</i> ; <i>Sphyrna lewini</i> ; <i>Sphyrna zygaena</i> ; <i>Taeniura grabata</i> ; <i>Torpedo torpedo</i> ; <i>Raja clavata</i> ; <i>Rostroraja alba</i> ; <i>Galeus polli</i> ; <i>Heptranchias perlo</i> ; <i>Squatina aculeata</i> ; <i>Torpedo nobiliana</i>
21	Shelf L	<i>Muraena helena</i> ; <i>Conger conger</i> ; <i>Lagocephalus laevigatus</i> ; <i>Fistularia tabacaria</i> ; <i>Dicentrarchus labrax</i> ; <i>Brotula barbata</i> ; <i>Zenopsis conchifer</i> ; <i>Zeus faber</i> ; <i>Ruvettus pretiosus</i> ; <i>Malacocephalus laevis</i> ; <i>Lophius budegassa</i> ; <i>Lophius piscatorius</i>
22	Shelf M	<i>Halobatrachus didactylus</i> ; <i>Ephippion guttifer</i> ; <i>Pseudupeneus prayensis</i> ; <i>Chelidonichthys lucerna</i> ; <i>Aspitrigla cuculus</i> ; <i>Trachinocephalus myops</i> ; <i>Uranoscopus polli</i> ; <i>Pomadasys incisus</i> ; <i>Pomadasys jubelini</i> ; <i>Pomadasys rogerii</i> ; <i>Dactylopterus volitans</i> ; <i>Acanthurus monroviae</i> ; <i>Callionymus lyra</i> ; <i>Mullus surmuletus</i> ; <i>Pomadasys perotaei</i> ; <i>Branchiostegus semifasciatus</i> ; <i>Scorpaena scrofa</i> ; <i>Gephyroberyx darwinii</i> ; <i>Pontinus accraensis</i> ; <i>Batrachoides didactylus</i> ; <i>Elops senegalensis</i> ; <i>Chloroscombrus chrysurus</i> ; <i>Alectis alexandrinus</i> ; <i>Selene dorsalis</i> ; <i>Campogramma glaycos</i> ; <i>Stromateus fiatola</i> ; <i>Caranx crysos</i> ; <i>Caranx senegalensis</i> ; <i>Trachinotus ovatus</i> ; <i>Seriola carpenteri</i> ; <i>Serranus cabrilla</i> ; <i>Cepola pauciradiata</i> ; <i>Scarus hoefleri</i> ; <i>Pontinus kuhlii</i> ; <i>Helicolenus dactylopterus dactylopterus</i> ; <i>Scorpaena stephanica</i>
23-24	Groupers	<i>Epinephelus aeneus</i> ; <i>Epinephelus caninus</i> ; <i>Epinephelus costae</i> ; <i>Epinephelus goreensis</i> ; <i>Epinephelus marginatus</i> ; <i>Epinephelus itajara</i> ; <i>Epinephelus fasciatus</i> ; <i>Mycteroperca rubra</i> ; <i>Rypticus saponaceus</i>
25-26	Sparids	<i>Pagellus acarne</i> ; <i>Pagellus bellottii bellottii</i> ; <i>Dentex maroccanus</i> ; <i>Dentex angolensis</i> ; <i>Dentex macrophthalmus</i> ; <i>Boops boops</i> ; <i>Sarpa salpa</i> ; <i>Spondylusoma cantharus</i>
27	Scianids	<i>Sciaena umbra</i> ; <i>Umbrina canariensis</i>
28	Shelf soles	<i>Arnoglossus imperialis</i> ; <i>Arnoglossus capensis</i> ; <i>Bothus podas</i> ; <i>Syacium guineensis</i> ; <i>Citharus linguatula</i> ; <i>Microchirus boscanion</i> ; <i>Monochirus hispidus</i> ; <i>Bathysolea polli</i> ; <i>Arnoglossus laterna</i> ; <i>Dicologlossa cuneata</i>
29	Shelf S	<i>Pterothrissus belloci</i> ; <i>Antennarius striatus</i> ; <i>Bembrops heterurus</i> ; <i>Saurida brasiliensis</i> ; <i>Serranus scriba</i> ; <i>Trachinus armatus</i> ; <i>Anthias anthias</i> ; <i>Capros aper</i> ; <i>Chaetodon hoefleri</i> ; <i>Trigloporus lastoviza</i> ; <i>Chelidonichthys obscurus</i> ; <i>Chilomycterus spinosus mauretanicus</i> ; <i>Chromis chromis</i> ; <i>Lepidotrigla cadmani</i> ; <i>Xyrichtys novacula</i> ; <i>Macroramphosus scolopax</i> ; <i>Chlorophthalmus agassizi</i> ; <i>Scorpaena notata</i> ; <i>Scorpaena angolensis</i> ; <i>Scorpaena laevis</i> ; <i>Synagrops microlepis</i> ; <i>Synchiropus phaeton</i> ; <i>Alepocephalus rostratus</i> ; <i>Nezumia sclerorhynchus</i>
30	Hake	<i>Merluccius senegalensis</i> ; <i>Merluccius polli</i>

Table S2. Diet composition for the base model

Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1 Marine mammals	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2 Coastal birds	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3 Meagre ad	0	0	0	0	0	0	0	0	0	0	00.10	0	0	0	0
4 Meagre juv	0	0.001	0	0	0	0	0	0	0	0	0	0	0	0.003	0
5 Mullet	4.81	2	7.69	8.33	0	0	0	0	0	0	5	0	0	0.096	0
6 Pelagic L	0	0.025	0	0	0	1.199	0	0	0	0	0	0	0	0.477	0
7 Mackerel	4.81	0	0	0	0	0.5	0	0	0	0	0	0	0	0	0
8 Sardine	4	0.654	0	0	0	18	3.0	0	0	0.10	2.4	0	0	4.077	0
9 Sardinella	12	18	15.39	8.33	0	49	8.9	0	0	1	15.2	2.0	0	7.075	1.25
10 Horse mackerels	0	0	0	0	0	7	1.98	0	0	0.21	4.5	1.0	0	1.075	1.25
11 Coastal selacians	0	0	0	0	0	0	0	0	0	0	0.18	0	0	0	0
12 Coastal M	4.81	3.5	15.00	8.33	0	0.12	0.17	0	0	0	1.5	0.82	0	3.824	0
13 Coastal S	4.81	3.4	2.85	0	0	3	1	0	0	1	5	6.1	0	2.069	1.25
14 Croakers ad	0	0.5	3.85	0	0	0	0	0	0	0	0.10	0	0	1.425	0
15 Croakers juv	0	0.2	0.10	8.33	0	0	0	0	0	0	0.03	0	0	0	0
16 Seabreams ad	0	0	16.70	0	0	0.48	0	0	0	0	3.0	0	0	1.075	0
17 Seabreams juv.	0	0.2	0.50	8.33	0	0	0	0	0	0	0.06	0.1	0	0	0
18 Catfish ad	0	0	0	0	0	0	0	0	0	0	0	0.5	0	0	0
19 Catfish juv	0	0.03	0	8.33	0	0	0	0	0	0	0	0.0	0	0	0
20 Shelf selacians	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21 Shelf L	0	0.01	0	0	0	0	0	0	0	0	0	0	0	0	0
22 Shelf M	4.81	0	7.69	8.33	0	0.24	0	0	0	0	1	0	0	2.417	0
23 Groupers ad	0	0	0	0	0	0	0	0	0	0	0.10	0.2	0	0.096	0
24 Grouper juv	0	0.0005	0	0	0	0	0	0	0	0	0.001	0	0	0	0
25 Sparids ad	0	1	3.30	0	0	1.319	0	0	0	0	0.50	0.4	0	1.171	0
26 Sparids juv	0	0.0005	0	0	0	0	0	0	0	0	0.09	0.05	0	0	0
27 Scianids	0	0.307	0	0	0	0.012	0	0	0	0	0	0	0	0.24	0
28 Shelf soles	4.81	2.4	3.85	8.33	0	0	0	0	0	0	0.10	0.10	0	1.306	0
29 Shelf S	4.84	2.917	0	0	0	6.2	0.25	0	0	5	9.5	3.6	0	1.120	0
30 Octopus vulgaris	0	0	0	0	0	0	0	0	0	0	0.45	0	0	0	0
31 Cephalopods	30.33	0.0001	7.69	4.167	0	0.7	2.0	0	0	0.20	3	0	0	1.438	0
32 BA L crustaceans	1.33	20	3.85	2.08	0	0	0.15	0	0	1.6	15	6	17	17.619	15.01
33 BA molluscs	0	20	0	0	6.44	0	0	0	0	0	2.1	7.60	1	0	0
34 BA worms	0	16	0	0	0	0	0.005	0	0	0	1.5	10.9	7	0	10.00
35 BA crustaceans	0	0.055	0	0	0	0	0.024	0	0	0.003	1.99	1.85	9	0	10.00
36 BA other inverts	0	0.1	0	0	0	0	0	0	0	0.014	0.09	2.00	0	0	0
37 BA meiobenthos	0	0.056	0	0	0	0	0	0	0	0	0	0	7.5	0	0
38 shelf L crustaceans	1.33	6.2	3.85	2.08	0	1	0.50	0	0	6	15.8	6.23	3	52.86	15.01
39 shelf molluscs	0	0	0	0	2.76	0.3	0	0	0	0	3.93	6.4	1	0	0
40 shelf worms	0	0	0	0	0	1.11	0.05	0	0	4.8	3.52	5.0	17.2	0	10.01
41 shelf crustaceans	0	0	0	0	0	0.8	0.24	0	0	0.02	4	2.09	12.77	0	10.01
42 shelf other inverts	0	0.035	0	0	0	0	0	0	0	0.05	0.09	4.75	0	0	0
43 shelf meiobenthos	0	0	0	0	0	0	0	0	0	0	0	0	7.5	0	0
44 mesozooplankton	0	0	0	5	0	0	55	90	66.76	35.5	0.07	2.4	7.33	0	10.2
45 macrozooplankton	0	0	3.85	7.5	0	7	17	0	0	15	0	1	1.18	0	3
46 BA mesozoopl.	0	0	0	5	0	0	2.4	0	11.1	6.50	0.07	2.7	7.33	0	10
47 BA macrozoopl.	0	0	3.85	7.5	0	0	2.13	0	0	15	0	4.68	1.18	0	3
48 BA phytoplankton	0	0	0	0	0	0	0	0	5.6	0	0	6.25	0	0	0
49 phytoplankton	0	0	0	0	0	0	0	10	16.7	0	0	1.56	0	0	0
50 algae and eelgrass	0	0	0	0	82.9	0	0	0	0	0	0	10.0	0	0	0
51 Detritus	0	0.422	0	0	7.9	0	5	0	0	1.956	0	3.71	0	0.523	0
52 Import	17.33	0.02	0	0	0	2.06	0.2	0	0	6.03	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0.17	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	1.00	0	0.40	3.62	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	4.08	0	0.20	0	0	0	0	0	0	0	0	1.50	0	0	0
8	0	0	0	0	0.09	7.78	2.40	1.86	0	0	0	0	0	0	0	5.0	0	0	0
9	0	0.5	0	0	0.09	9.00	4.21	1.86	3.82	1.39	0	0	0	0	1	15.70	0	0	0
10	0	0	3.3	0	3.4	10.50	0.30	1.10	3.82	0	0	1.33	0	0	1	8.6	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0.2	0	1	0	2.02	0.38	0.20	3.07	0	0.69	0	1.56	0.95	0	0	0	0	0	0
13	4.7	0.1	7	0	0.17	1.45	4.38	13.3	3.82	1.26	0	1.78	0.95	0.10	0	1.7	0	0	0
14	0	0	0.4	0	0.17	0.14	0	0	0	0	0	0	0	0	0	0	0	0	0
15	0	0.1	0	0	0	0	0	0	3.82	0	0	0	0	0	0	0	0	0	0
16	0	0	10	0	3.57	2.00	0.50	1.38	0	0	0	0.78	0	0	0	0	0	0	0
17	0	0	0.04	0	0	0	0	0	3.82	0.02	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
19	0	0	0	0	0	0	0	0	3.82	0	0	0	0	0	0	0	0	0	0
20	0	0	0	0	0.83	2.00	0	0	0	0	0	0	0	0	0	0	0	0	0
21	0	0	0	0	2.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0
22	0	0	4	0	5.55	20.0	1.50	2.20	0	0	0	2.68	0	0	0	0.5	0	0	0
23	0	0	0	0	1.2	0.14	0	0	0	0	0	0.67	0	0	0	0	0	0	0
24	0	0	0	0	0	0.005	0	0	0	0	0	0	0	0	0	0	0	0	0
25	0	0	1.6	0	3.23	3.00	0	1.10	0	0	0	0.67	1.89	0	0	2	0	0	0
26	0	0	0.1	0	0	0.10	0	0	3.82	0	0	0.1	0	0	0	0.01	0	0	0
27	0	0	0.3	0	1.2	1.53	0	0.2	0	0	0	0	0	0	0	0	0	0	0
28	0.5	0	1	0	1.2	1.74	0.20	0.48	3.82	0	0	0	1.89	0	0	0	0	0	0
29	5.0	1	4.7	0	9.87	27.0	3.00	3.70	3.82	1.49	0	2.9	1.89	0.10	0	5.0	0	0	0
30	0.6	0.07	0	0	0.03	0	0	7.19	0	4.07	0	0	0	0	0	4	0	0	0
31	0.2	0.07	0	0	15.3	1.27	0	1.25	0	4.07	4.54	0	0	0.15	0.5	1.00	0	0	0
32	6	10	23	0	2.31	0	9.5	134	16.7	1.76	0	12.5	9	3	10	9	5	0	0
33	10	15	3	20	0	0	2.8	0.03	16.15	1.81	4.54	0	1.3	1.9	2	0	9	5	0
34	17.7	15	3	15	0.37	0	2	0	0	5.5	4.54	4.0	1	5.5	2	1	9	0	0
35	1.0	13	0	15	0.04	0	0.50	0	0	0.15	4.54	0.6	2.49	0.20	0	0.10	1.5	0	0
36	0.5	0.002	0	0	0	0	0.10	0.41	0	1.5	4.54	3.21	0	0.10	0	0	1	0	0
37	0.02	0	0	0	0.37	0	0	0	0	4.69	4.54	1.56	0.39	3.20	0	0	14.5	0	3
38	5.0	7.7	5	0	16.3	5.54	5.00	42	16.65	5.28	0	37.49	29.74	1.5	15	10.00	0	0	0
39	25.0	13.8	10	20	0	0	10.1	0.08	16.15	5.44	4.54	0	13.19	13	51.5	0	0	0	0
40	4.9	10	14	15	1.12	0	22.8	0	0	2.36	4.54	9.37	14.95	29.2	17	29.5	0	0	0
41	6.2	13.7	0	15	0.13	0.20	4.56	0	0	0.44	4.54	4.69	12.24	9	0	0.40	0	0	0
42	7.1	0.002	0	0	0	0.004	6.56	1.23	0	18.4	4.54	9.37	2.38	4	0	0	0	0	0
43	0.02	0	0	0	1.12	0	0	0	0	14.07	4.54	4.69	1.16	11.81	0	0	0	0	0
44	0	0	0	0	0	0.02	10.8	0	0	0	0	0	2	14.5	0	0	0	0	0
45	0	0	3.3	0	1.12	0.41	1	0	0	0	0	0	2	2	0	4.00	0	0	0
46	0	0	0	0	0	0	2.47	0	0	0	0	0	0	0.40	0	0	0	0	0
47	0	0	0	0	0.37	0	1.46	0	0	0	0	0	0.53	0.2	0	1.00	0	0	0
48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0
49	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
50	5.2	0	0	0	0	0	0	0	0	25.57	50	0	0	0	0	0	20	0	0
51	0	0	5	0	0	0	3.01	0	0	0	0	0	0	0	0	0	40	85	97
52	0	0	0	0	21.1	5.8	0	0	0	0	0	0	0	0.12	0	0	0	0	0

Table S3. Effort used for each fleet

Year	Artisanal N boats ^a	Industrial demersal N boats ^a	Industrial pelagic mean GRT ^b
1991	742	160	2703
1992	681	190	3223
1993	1217	219	3630
1994	1540	251	4501
1995	1986	317	4730
1996	2587	318	4407
1997	2728	327	4200
1998	3142	312	4523
1999	2640	289	4704
2000	2750	267	5087
2001	2850	335	5002
2002	3700	357	4882
2003	3800	334	4980
2004	3950	328	4898
2005	3950	302	5003
2006	3950	302	5148

^a. [6]^b. [16]

Table S4. Composition of the marine mammal group and parameters values for biomass, and P/B and Q/B ratios

English name	Latin name	Biomass t/km ²	P/B /year	Q/B /year ^e
Killer whale	<i>Orcinus orca</i>	0.0026	^a 0.02 ^c	7.58
Monk seal	<i>Monachus monachus</i>	0.0020	^b 0.05 ^d	12.11
Harbour porpoise	<i>Phocoena phocoena</i>			14.18
Dolphins, pilot whales, and pigmy sperm whale	<i>Sousa teuszii</i> , <i>Grampus griseus</i> , <i>Tursiops truncatus</i> , <i>Stenella frontalis</i> , <i>Stenella attenuata</i> , <i>Stenella</i> <i>longirostris</i> , <i>Stenella coeruleoalba</i> , <i>Stenella clymene</i> , <i>Delphinus delphis</i> , <i>Globicephala melas</i> , <i>Globicephala</i> <i>macrorhincus</i> , <i>Kogia breviceps</i>	0.0001	^a 0.036 ^a	9.92
Beaked whales	<i>Mesoplodon densirostris</i> , <i>M.</i> <i>europaeus</i> , <i>Ziphius cavirostris</i>	0.0001	^a 0.036 ^a	9.92
Weighted average/sum		0.0076	0.05	12.56

^a. [17]^b. [18]^c. [19]^d. the P/B is assumed equal to that of dolphins (0.05)^e. Calculated from ration (*R*) in % of body weight per day [19]: $R=0.1*W^{0.8}$ where *W* is the body weight in kg taken from [50]

Table S5. Composition of the coastal bird groups and parameters used in the model.

BW is body weight

Latin name	Common name	N pairs	N summer	N winter	residence	BW (kg)	Biom. (t)	QB /year	M /year
Breeders									
<i>Pelecanus onocrotalus</i>	White pelican	3080 ^a	9240		12 ^f	10.50 ^b	97	46	
<i>Phoenicopterus ruber</i>	Greater flamingo	12940 ^a	38820		6 ^f	3.00 ^c	58	56	
<i>Microcarbo africanus</i>	Long-tailed cormorant	2460 ^a	7380		6 ^f	0.62 ^c	2	71	0.10 ^v
<i>Phalacrocorax carbo lucidus</i>	Cormorant	4260 ^a	12780		12 ^f	2.25	29	58	
<i>Platalea leucorodia</i>	Spoonbill	1610 ^a	4830		12 ^f	1.36 ^c	7	63	0.19 ^w
<i>Ardea cinera</i>	Grey heron	2400 ^a	7200		12 ^f	0.90 ^c	6	67	0.361
<i>Egretta garzetta/gularis</i>	Little egret/reef heron	745 ^a	2235		12 ^f	0.50 ^c	1	73	
<i>Larus genei</i>	Slender-billed gull	1610 ^a	4830		12 ^f	0.30 ^c	1	79	
<i>Chroicocephalus cirrocephalus</i>	Grey-headed gull	15 ^a	45		12 ^f		0		
<i>Gelochiledon nilotica</i>	Gull-billed tern	1180 ^a	3540		12 ^f	0.27 ^x	1		
<i>Hydroprogne caspia</i>	Caspian tern	2575 ^a	7725		12 ^f	0.65 ^c	5	70	0.131
<i>Thalasseus maximus</i>	Royal tern	5630 ^a	16890		12 ^f	0.40 ^c	7	76	
<i>Sterna albifrons</i>	Little tern	30 ^a	90		12 ^f	0.06 ^c	0	101	
<i>Sterna hirundo</i>	Common tern	100 ^a	300		6 ^f	0.12 ^c	0	91	0.321
<i>Onychoprion anaethetus</i>	Bridled tern	440 ^a	1320		8 ^b	0.12 ^x	0		
total breeders							215	54	0.22
Waders									
<i>Haematopus ostralegus</i>	Oystercatcher			129 ^{r,s}	8 ^b	0.55 ^c	1	72	
<i>Charadrius hiaticula</i>	Ringed plover			1805 ^{r,s}	8 ^b	0.05 ^b	2	104	
<i>Charadrius alexandrinus</i>	Kentish plover			85 ^{r,s}	8 ^b	0.04 ^b	0	108	0.221
<i>Pluvialis squatarola</i>	Grey plover			457 ^{r,s}	8 ^b	0.19 ^b	1	85	
<i>Calidris canutus</i>	Knot			26635 ^{r,s}	8 ^b	0.12 ^b	54	91	
<i>Calidris alba?</i>	Sanderling			484 ^{r,s}	8 ^b	0.05 ^b	0	104	
<i>Calidris minuta</i>	Little stint			700 ^{r,s}	8 ^b	0.02 ^b	0	117	
<i>Calidris ferruginea</i>	Curlew Sandpiper			7770 ^{r,s}	8 ^b	0.05 ^b	7	103	
<i>Calidris alpina</i>	Dunlin			28936 ^{r,s}	8 ^b	0.04 ^b	21	106	
<i>Limosa lapponica</i>	Bar-tailed godwit			5756 ^{r,s}	8 ^b	0.03 ^b	2	114	
<i>Numenius phaeopus</i>	Whimbrel			1048 ^{r,s}	8 ^b	0.41 ^b	7	75	
<i>Numenius arquata</i>	Curlew			213 ^{r,s}	8 ^b	0.72 ^b	3	69	
<i>Tringa totanus</i>	Redshanks			4765 ^{r,s}	8 ^b	0.11 ^b	9	92	0.291
<i>Tringa nebularia</i>	Greenshank			88 ^{r,s}	8 ^b	0.18 ^b	0	85	
<i>Arenaria interpres</i>	Turnstone			408 ^{r,s}	8 ^b	0.10 ^b	1	93	
total waders							109	93	0.46
coastal birds							324	67	0.30

^a Campredon 1987 (for year 1984-85) in [23]. Npairs* 3 to estimate the whole population (Nsummer)^b [23]^c [51]; May 1988 shelf edge bordering BA; 20->1000m^d count on Feb 1997, [24].^e <http://www.oiseaux.net/oiseaux/>^f based on [52]; species that are present in both winter and summer are assumed to be present 12 months, averaging abundances^g assumed^h preparing to migrate in May so assumed to have spent winter in BAⁱ start arriving in Oct, mainly present Dec-Feb^j winters further south, migrate through from March to May^k [53]^l from survival estimate from [27]^m [54]ⁿ based on Leach's storm petrel minimum as weight is max weight from sept-nov in Mauritania [55]^p [56]^q [57]^r [24]^s original counts in Baie Arguin; multiplied by 25.5 to account for surface in the rest of the Banc Arguin^t present in the fall and in May^u assumed equal to common tern^v from the survival value for adults in [26] for unspecified cormorants^w from the survival value for adults in [28]^x average body weights of other terns^y obtained from the equation Log(Ration)=-0.293+0.85*Log(BW) [25] where BW is the body weight in kg.

Table S6. Parameters used to calculate benthos biomass and production by unit of biomass (P/B).

AFDW: Ash-free dry weight; DW: dry weight; WW: wet-weight; surf adj: adjusted for surface ratio;

Group	AFDW/ WM	Banc d'Arguin			Biomass shelf					P/B /year
		AFDW g/m ²	WW g/m ²	surf adj	AFDW /g/m ²	N 106/m ² m	body weight (micro g)	DW/ WW	WW g/m ²	
Crustaceans			0.79	0.79 ^v	1.49 ^{j,k}				9.92	2.4 ^q
Amphipoda	0.15 ^a	0.0348 ^b	0.23							
Crustacea	0.15 ^a	0.0022 ^b	0.01							
Isopoda	0.15 ^a	0.0479 ^b	0.32							
Sipunculida	0.15 ^a	0.0342 ^b	0.23							
Tanaidacea	0.15 ^a	0.0007 ^b	0.00							
Molluscs			132.55	94.00	4.84 ^j				32.36	1.5
Gastropoda	0.16 ^c	1.8521 ^b	11.36	11.36 ^u						1.5
Bivalvia	0.14 ^c	10.981 ^b	80.74	80.74 ^u						1.5 ^r
Senilia senilis	0.14 ^c	8.1000 ^b	40.44	1.90 ^v						0.02
Worms			28.00	28.00 ^u					39.22	3 ^q
Nemertinea	0.17 ^c	0.0132 ^b	0.08	0.08						
Oligocheta	0.05 ^c	0.1044 ^b	1.90	1.90	0.96 ^{j,l}				0.05	
Planaria	0.05 ^f	0.0019 ^b	0.03	0.03						
Polychaeta	0.15 ^c	3.8972 ^b	25.98	25.98	5.88 ^j				39.17	
Other inverts			0.55	0.55 ^v					21.25	
Coelenterata	0.12 ^{c,d}	0.0019 ^b	0.02							
Echinodermata	0.11 ^{c,e}	0.0581 ^b	0.52							
Holothuroidea	0.077 ^c									
Ophiuroidea	0.148 ^c				3.14 ^j				21.20	1.8 ^q
Insecta	0.16 ^c	0.0023 ^b	0.01							
Miscellaneous					0.45 ^s				0.05	
Total macrobenthos		17.031	161.89	123.34	16.8				196.3	
Meiobenthos			9.07	6.91					11.0	9 ^p
Meiobenthos			9.07	6.91 ^h						
Nematodes						2.56 ^j	0.453 ⁿ	0.25	4.6	
Copepods						0.41 ^j	1.90 ^o	0.214	3.7	
Foraminiferans							DW			
Crustaceans L				51.67		2.70 ^j	1 WW ^p	na	2.7	
<i>Uca tangeri</i>	0.75 ^{c, g}	0.589 ⁱ	1.27	0.06						
Shrimp	^t		51.61	51.61						

^a value used for crustaceans by [41]

^b Values for tidal flats from Table 3 in [41]; the surface of tidal flats is assumed to be 500 km² [42]

^c from [58], molluscs conversion without shell

^d using Actinaria as an example

^e average of holothurians and Ophiuridae based on species list

^f assumed equal to that of Oligocheta

^g based on *Uca lactea*

^h assumed that meiobenthos biomass was in the same proportion as the ratio of meiobenthos/macrobenthos as on the shelf. By comparison, Wolff et al [41] estimate that 3-10% of benthos is not retained in their sieve

ⁱ 0.5-1 g AFDW/m² [41]

^j [40]

^k Isopoda, Cumacea, Gammaridae, Tanaidacea

^l assumed to be mainly Oligochaeta and that the same density of other inverts were present on the shelf as on tidal flats

^m [40]

ⁿ [59]; St Lawrence River, Canada. coastal nematodes are larger than deep-sea; 0.44 and 0.37 microg in Cretan Sea [60]

^o based on average of size range observed in figure 1 in [61]

^p assumed same value as in [45]

^q based on empirical equation [44] which yielded similar results as that of [43]

^r assumed value

^s assumed same density of other invertebrates as on tidal flats

^t push net sampling with correction for under-sampling [42]. Nevertheless, *Penaeus* sp are probably underestimated as well as individuals smaller than 10 mm. Assumed individual body weight of 0.5 g for *Hippolyte inermis* and 1 g for *Palaemon* spp. and *Penaeus* spp.

^u Biomass for the entire Banc = Btidal (P+0.5(1-P)) where P is the proportion of the tidal flats in the Banc d'Arguin = 0.05, and assuming that the biomass was half as abundant in subtidal habitats

Table S7. List of biomass and catch time series available and used in the Ecosim fitting process.

n/a: not available

	Group name	Time series	
		Biomass	Catch
1	Marine mammals	n/a	n/a
2	Coastal birds	n/a	n/a
3	Meagre ad	n/a	available
4	Meagre juv	n/a	n/a
5	Mulletts	n/a	available
6	Pelagic L	n/a	available
7	Mackerel	n/a	available
8	Sardine	not used	available
9	Sardinelles	available	available
10	Horse mackerels	not used	available
11	Coastal selacians	available	available
12	Coastal M	available	available
13	Coastal S	n/a	n/a
14	Croakers ad	available	available
15	Croakers juv		n/a
16	Seabreams ad	available	available
17	Seabreams juv.	n/a	n/a
18	Catfish ad	not used	available
19	Catfish juv	n/a	n/a
20	Shelf selacians	available	available
21	Shelf L	available	available
22	Shelf M	available	available
23	Groupers ad	available	available
24	Grouper juv	n/a	n/a
25	Sparids ad	available	available
26	Sparids juv	n/a	n/a
27	Scianids	available	available
28	Shelf soles	available	available
29	Shelf S	available	available
30	Octopus vulgaris	available	available
31	Cephalopods	available	available
32	BA L crustaceans	available	n/a
33	BA molluscs	n/a	n/a
34	BA worms	n/a	n/a
35	BA crustaceans	n/a	n/a
36	BA other inverts	n/a	n/a
37	BA meiobenthos	n/a	n/a
38	shelf L crustaceans.	n/a	available
39	shelf molluscs	n/a	available
40	shelf worms	n/a	n/a
41	shelf crustaceans	n/a	n/a
42	shelf other inverts	n/a	n/a
43	shelf meiobenthos	n/a	n/a
44	mesozooplankton	n/a	n/a
45	macrozooplankton	n/a	n/a
46	BA mesozoopl.	n/a	n/a
47	BA macrozoopl.	n/a	n/a
48	BA phytoplankton	n/a	n/a
49	phytoplankton	n/a	n/a
50	algae and eelgrass	n/a	n/a
51	Detritus	n/a	n/a

Table S8. Proportion of invertebrates from the Banc (pBAi) in diets imposed in each model (M30, Base and P30), and resulting biomass, ecotrophic efficiency (EE) from balancing the Ecopath model, and vulnerability values for each Ecosim model (M30, Base and P30) fitted to the time series.

Only vulnerability values different from the default value (2) are listed. Values in bold are biomass and EE that were calculated by Ecopath instead of being input.

	Group name	Balanced Ecopath model									Vulnerability (Ecosim)		
		pBAi (imposed)			M30		Base		P30		M30	Base	P30
		M30	Base	P30	Biom	EE	Biom	EE	Biom	EE			
1	Marine mammals	0.5	0.5	0.5	0.01	0	0.01	0	0.01	0			
2	Coastal birds	1	1	1	0.01	0	0.01	0	0.01	0			
	Meagre												
3	Meagre ad	0.33	0.5	0.75	0.12	0.66	0.12	0.66	0.12	0.66	1.002	1.01	1.004
4	Meagre juv	0.33	0.5	1	5E-05	0.97	5.3E-05	0.97	5E-05	0.97			
5	Mulletts	0.47	0.7	1	0.42	0.8	0.42	0.80	0.42	0.8	1	1	1
6	Pelagic L	0	0	0	3.42	0.9	3.42	0.90	3.89	0.8	1.713	7.205	4.473
7	Mackerel	0.1	0.1	0.1	1.45	0.74	1.45	0.74	1.45	0.75	1	1.002	1.05
8	Sardine	0	0	0	11.79	0.77	11.79	0.77	11.79	0.83	1	2	2
9	Sardinelles	0.25	0.25	0.25	18	0.78	18	0.78	18	0.85	1	1	1
10	Horse mackerels	0.25	0.25	0.25	10	0.84	10	0.84	10	0.87	1	2.144	2.002
11	Coastal selacians	0.33	0.5	0.75	1.24	0.01	1.24	0.01	1.24	0.01	2.338	2.265	2.365
12	Coastal M	0.33	0.5	0.75	0.83	0.86	0.83	0.86	0.83	0.86	1.12	1.11	1.179
13	Coastal S	0.33	0.5	0.75	4.21	0.95	4.21	0.95	5.18	0.8	1.477	>100	>100
	Croakers												
14	Croakers ad	0.17	0.25	0.375	0.08	0.75	0.077	0.75	0.08	0.75	1.101	1	1
15	Croakers juv	0.33	0.5	1	0.00	0.71	0.00352	0.71	0.00	0.71			
	Seabreams												
16	Seabreams ad	0.33	0.5	0.75	1.69	0.88	1.69	0.88	1.69	0.90	1.574	1.838	2.598
17	Seabreams juv.	0.33	0.5	1	0.01	0.88	0.0125	0.88	0.01	0.88			
	Catfish												
18	Catfish ad	0.33	0.5	0.75	0.60	0.23	0.6	0.23	0.60	0.23	1	1	1.005
19	Catfish juv	0.33	0.5	1	0.00	0.78	0.00166	0.78	0.00	0.78			
20	Shelf selacians	0.25	0.25	0.25	0.20	0.79	0.2	0.79	0.20	0.79	1	1	1
21	Shelf L	0	0	0	0.36	0.47	0.36	0.47	0.36	0.47	1	1	1
22	Shelf M	0.25	0.25	0.25	1.55	0.91	1.55	0.91	1.55	0.91	1	1	1.048
	Groupers												
23	Groupers ad	0.25	0.25	0.25	0.11	0.94	0.11	0.94	0.11	0.94	1.03	1.039	1.037
24	Grouper juv	0.33	0.5	1	0.00	0.54	0.00037	0.54	0.00	0.54			
	Sparids												
25	Sparids ad	0.25	0.25	0.25	1.29	0.87	1.29	0.87	1.29	0.93	1	1	1
26	Sparids juv	0.33	0.5	1	0.011	0.94	0.01074	0.94	0.01	0.94			
27	Scianids	0.25	0.25	0.25	0.22	0.67	0.22	0.67	0.22	0.67	>100	>100	>100
28	Shelf soles	0.25	0.25	0.25	0.35	0.88	0.35	0.88	0.35	0.88	1	1	1
29	Shelf S	0.25	0.25	0.25	6.19	0.95	6.19	0.95	7.61	0.8	21.02	>100	>100
30	Octopus vulgaris	0.2	0.2	0.2	1.37	0.63	1.37	0.63	1.37	0.63	4.45	1.116	1.132
31	Cephalopods	0.2	0.2	0.2	1	0.84	1	0.84	1	0.87	2	4.165	4.065
32	BA L crustaceans	1	1	1	7.66	0.88	9.12	0.88	11.02	0.83			
33	BA molluscs	1	1	1	14.01	0.93	17.86	0.89	22.86	0.91			
34	BA worms	1	1	1	4.15	0.90	5.32	0.81	6.84	0.86			
35	BA crustaceans	1	1	1	1.14	0.87	1.14	0.98	2.09	0.98			
36	BA other inverts	1	1	1	0.57	0.93	0.57	0.87	0.95	0.90			
37	BA meiobenthos	1	1	1	2.09	0.77	2.09	0.93	2.66	0.93			
38	shelf L crustac.	0	0	0	8.91	0.83	8.10	0.73	8.10	0.74	2	1.966	2.061
39	shelf molluscs	0	0	0	26.21	0.51	26.21	0.50	26.21	0.49			

		Balanced Ecopath model									Vulnerability (Ecosim)		
		pBAi (imposed)			M30		Base		P30				
	Group name	M30	Base	P30	Biom	EE	Biom	EE	Biom	EE	M30	Base	P30
40	shelf worms	0	0	0	31.77	0.43	31.77	0.41	31.77	0.43			
41	shelf crustaceans	0	0	0	8.04	0.83	8.04	0.80	9.04	0.78			
42	shelf other inverts	0	0	0	17.21	0.23	17.21	0.22	17.21	0.22			
43	shelf meiobenthos	0	0	0	8.91	0.27	8.91	0.25	8.91	0.26			
44	mesozooplankton	0	0	0	55.08	0.15	55.08	0.15	55.08	0.15			
45	macrozooplankton	0	0	0	3.41	0.73	3.41	0.73	3.41	0.75			
46	BA mesozoopl.	1	1	1	1.75	0.8	1.78	0.8	1.91	0.8			
47	BA macrozoopl.	1	1	1	2.46	0.8	2.50	0.8	2.59	0.8			
48	BA phytoplankton				6	0.38	6	0.40	6	0.44			
49	phytoplankton				68	0.85	68	0.85	68	0.85			
50	algae and eelgrass				549	0.01	549	0.01	549	0.01			
51	Detritus				560	0.45	560	0.46	560	0.48			

Table S9. Biomass, catch and fishing mortality (C/B) by functional groups based on EwE data (1991) and estimations (2006).

no	Group name	Biomass 1991	Biomass 2006	Catch 1991	Catch 2006	C/B 1991	C/B 2006
1	Marine mammals	0.009999	0.008541				
2	Coastal birds	0.010001	0.010549				
3	Meagre ad	0.12004	0.059078	0.014005	0.024851	0.12	0.42
4	Meagre juv	5.34E-05	3.35E-05				
5	Mullet	0.4206	0.339271	0.103147	0.268065	0.25	0.79
6	Pelagic L	3.420516	1.627229	2.679404	1.333773	0.78	0.82
7	Mackerel	1.450124	1.656654	0.289025	0.341271	0.20	0.21
8	Sardine	11.79149	17.73109	1.801227	2.799447	0.15	0.16
9	Sardinella	18.0042	19.65045	2.310539	3.112236	0.13	0.16
10	Horse mackerels	10.00185	11.97514	3.812705	4.733549	0.38	0.40
11	Coastal selacians	1.208741	0.343718	0.061412	0.076773	0.05	0.22
12	Coastal M	0.830443	0.601779	0.121065	0.297622	0.15	0.49
13	Coastal S	4.211359	6.68667				
14	Croakers ad	0.077201	0.070635	0.003008	0.008346	0.04	0.12
15	Croakers juv	0.003531	0.002785				
16	Seabreams ad	1.69036	1.021793	0.167036	0.281987	0.10	0.28
17	Seabreams juv.	0.012507	0.007336				
18	Catfish ad	0.600135	0.490796	0.034008	0.148055	0.06	0.30
19	Catfish juv	0.001666	0.001246				
20	Shelf selacians	0.199917	0.17086	0.010995	0.020673	0.06	0.12
21	Shelf L	0.360019	0.487229	0.071004	0.181375	0.20	0.37
22	Shelf M	1.550139	1.424855	0.146013	0.284911	0.09	0.20
23	Groupers ad	0.107764	0.018469	0.024492	0.01773	0.23	0.96
24	Grouper juv	0.000367	0.000103				
25	Sparids ad	1.290003	1.121018	0.014	0.049836	0.01	0.04
26	Sparids juv	0.010753	0.009952				
27	Scianids	0.214149	0.033206	0.016548	0.005362	0.08	0.16
28	Shelf soles	0.349963	0.25992	0.008999	0.015167	0.03	0.06
29	Shelf S	6.197166	13.73088	0.005002	0.020918	0.00	0.00
30	Octopus vulgaris	1.35745	0.384884	0.874911	0.678659	0.64	1.76
31	Cephalopods	0.985809	0.31217	0.251381	0.151324	0.26	0.48
32	BA L crustaceans	9.124096	5.993826				
33	BA molluscs	17.86009	19.3619				
34	BA worms	5.319961	4.142092				
35	BA crustaceans	1.140169	0.502134				
36	BA other inverts	0.570191	0.661913				
37	BA meiobenthos	2.08967	2.376096				
38	shelf L crustaceans	8.106485	6.285361	0.005004	0.009989	0.000617	0.001589
39	shelf molluscs	26.21321	20.85251	3.01E-06	4.52E-06	1.15E-07	2.17E-07
40	shelf worms	31.77156	24.25814				
41	shelf crustaceans	8.035142	3.894845				
42	shelf other inverts	17.21303	15.76241				
43	shelf meiobenthos	8.909588	8.506203				
44	mesozooplankton	55.07933	54.8538				
45	macrozooplankton	3.410281	3.048343				
46	BA mesozooplankton	1.779772	1.793211				
47	BA macrozooplankton	2.500826	2.286842				
48	BA phytoplankton	5.886206	5.880416				
49	phytoplankton	67.82123	68.7425				

Figure 1 is a trophic network diagram showing the relationships between various species across four trophic levels (1 to 4) and three habitats (Coast, Shelf, Pelagic). The diagram uses different line styles and colors to represent various types of interactions: solid black lines for predation, dashed black lines for predation and competition, blue dashed lines for indirect impact, and red dotted lines with '+' for positive impact. Species are grouped by habitat and trophic level. Key species include mammals, birds, meagJ, meagA, croakA, croakJ, catA, seabJ, catJ, sparidsJ, mullets, grouA, grouJ, scian, octopus, sparA, shelf benthos, cephal, Shelf L, Shelf sel, Shelf S, hnmack, sardine, sardinelle, mackerel, mesozoopl, and phytoplankton. The diagram illustrates a complex web of interactions, with many species having multiple predators and prey. The legend at the bottom clarifies the line styles and colors used.

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